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# GreenSnap AI: An AI-enabled geo-verified waste reporting and verification system

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ABSTRACT

Urban waste mismanagement remains a serious environmental and public-health challenge, particularly in rapidly growing urban and semi-urban regions. This paper presents GreenSnap-AI, an open-source mobile and web-based software platform for AI-assisted, geo-verified waste reporting and cleanup verification. The system enables citizens to submit geo-tagged waste images through a React Native mobile application, while supervisors and administrators manage verification, cleanup, and monitoring through role-specific mobile and web interfaces. The backend, implemented using Node.js/Express and MongoDB, coordinates user authentication, report management, AI-based image validation, and Haversine-based geospatial verification. Image validation is performed through decoupled AI inference services using MobileNetV3-Large and YOLOv11 models, allowing model replacement without major changes to the core application logic. A 10-meter geo-fence is used to verify that cleanup actions are performed at the reported location. Evaluation on a custom binary waste-presence verification dataset containing 17,556 labeled images using a held-out test subset demonstrated strong performance, with YOLOv11 achieving approximately 99.5% accuracy under the defined evaluation protocol. This result reflects the effectiveness of the model for binary waste validation within the evaluated deployment scenario rather than fine-grained waste recognition. Field testing in Sagri Village, Rawalpindi, demonstrated the practical feasibility of the platform for citizen-led reporting and municipal supervision. The main contribution of GreenSnap-AI is not limited to image classification; rather, it provides a reusable software workflow that integrates visual validation, spatial accountability, role-based access control, and administrative monitoring for transparent waste-management operations.

Meta data.

| Nr | Code metadata description | *Metadata* |
|---|---|---|
| C1 | Current code version | V1.0 |
| C2 | Permanent link to code/ repository used for this code version | https://github.com/Muhammad5Ali/GS_Web_App |
| C3 | Legal code license | *MIT* |
| C4 | Code versioning system used | Git-based version control |
| C5 | Software code languages, tools and services used | *JavaScript (React Native, React.js, Node.js/ Express), MongoDB, Leaflet.js, Python (for AI microservices), TensorFlow, Ultralytics YOLO v11, MobileNetV3* |
| C6 | Compilation requirements, operating environments and dependencies | *Android/iOS (via Expo) for mobile app; Web browsers (modern) for admin portal; Node.js (v14+) server; MongoDB (v4+); Internet connectivity for cloud ML inference; third-party services: Hugging Face Spaces and Ultralytics Hub APIs.* |
| C8 | Support email for questions | mohammad.bscs4490@iiu.edu.pk \|\| rajaali8383679@gmail.com |

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## 1. Motivation and significance

Rapid urban growth has increased the volume and complexity of municipal solid waste, creating serious environmental, public-health, and administrative challenges [1]. In many developing regions, including Pakistan, waste collection is often affected by delayed reporting, limited monitoring capacity, weak accountability mechanisms, and the absence of reliable digital evidence for cleanup verification [2,3]. While policy reforms and municipal investment remain essential, software-based civic reporting systems can strengthen transparency by enabling citizens to report waste incidents, supporting authorities with real-time geospatial data, and creating auditable records of cleanup activities. GreenSnap-AI addresses this need through an integrated software platform. The platform combines citizen reporting, AI-assisted image validation, role-based supervision, and location-based cleanup verification [4].

GreenSnap-AI combines AI-based waste-image validation with geospatial verification to support accountable citizen reporting. Citizens submit geo-tagged waste reports through a mobile application, while cleanup activities are verified using both visual evidence and a 10-meter geo-fence constraint.

Unlike existing smart waste monitoring and citizen-reporting platforms that primarily focus on waste detection, complaint submission, or location visualization, GreenSnap-AI introduces an enforceable verification workflow where both visual and spatial evidence are required before a cleanup action can be finalized. The contribution of this work is not a new deep learning architecture or geospatial algorithm; instead, it is a software-level innovation that integrates established AI inference models, geospatial validation mechanisms, and multi-stakeholder workflows into a unified and auditable civic monitoring platform. By connecting citizen reporting, AI-assisted validation, supervisor verification, and administrative oversight within a closed-loop workflow, GreenSnap-AI transforms conventional reporting systems into an accountable cleanup management process.

The platform's contributions include:

- Geo-enforced AI validation for waste reporting: The proposed framework combines AI-based waste recognition with mandatory spatial verification, ensuring that submitted reports satisfy both visual and location-based validation requirements [5,6].
- Real-time cleanup verification: A Haversine-based geo-fencing mechanism verifies that cleanup activities are performed at the reported location, extending geospatial validation concepts to practical municipal waste-management workflows [7,8].
- End-to-end civic software platform: GreenSnap-AI integrates a React Native mobile application, React.js administrative dashboard, Node.js/Express backend, and MongoDB data layer within a modular MVC architecture that supports scalability and maintainability (Fig. 1).
- Experimental and field validation: The framework was evaluated using a large waste-image dataset and a real-world municipal deployment, demonstrating improved reliability compared with vision-only reporting approaches.

Overall, GreenSnap-AI provides a reusable software model for citizen-assisted municipal waste reporting. Its main contribution is the integration of AI-based visual validation and Haversine-based geospatial verification into a single operational workflow. This closed-loop design supports transparent reporting, supervisor accountability, and administrative monitoring in practical waste-management environments.

The remainder of this paper is organized as follows. Section 1.1 reviews related work. Section 2 describes the software design, architecture, functionalities, and evaluation results, including experimental validation and field testing. Section 3 discusses the impact of the software. Section 4 concludes the paper and outlines future work.

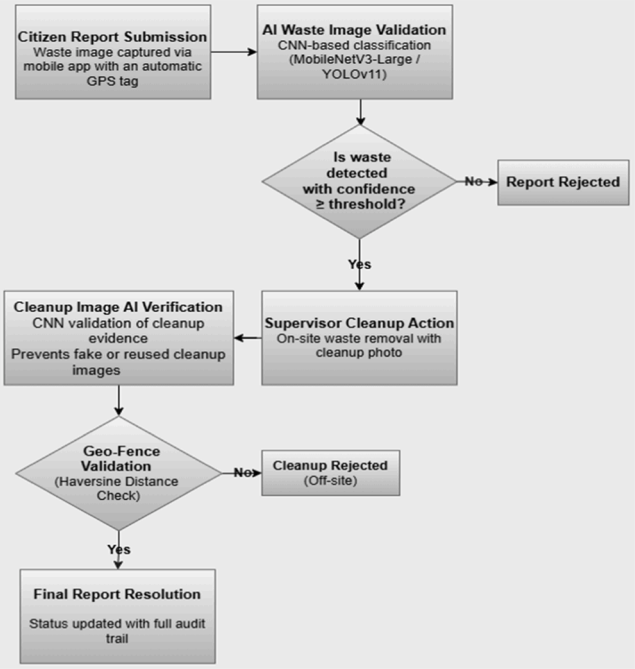



**Fig. 1.** Geo-verified AI decision workflow of GreenSnap-AI showing the interaction between citizen reporting, AI-based image validation, supervisor cleanup verification, Haversine-based geo-fence checking, and final report status updating.

### 1.1. Related work

Existing research on smart waste-management systems can be broadly categorized into AI-based waste detection, citizen-centered environmental reporting, and geospatial verification approaches. Recent deep learning-based methods have demonstrated improved waste recognition capability using lightweight convolutional architectures and YOLO-based detectors, mainly focusing on classification accuracy, object detection performance, and real-time inference efficiency [5,6,9–19]. However, these approaches primarily address visual recognition and do not incorporate mechanisms to verify whether reported waste or completed cleanup activities correspond to the actual geographic location [20–22].

Parallel research on civic reporting platforms has explored mobile-based citizen participation and digital environmental monitoring, enabling users to submit location-aware reports and support municipal decision-making [23–29]. However, many existing platforms depend on manual inspection or provide location information without enforcing spatial validation. Geospatial verification techniques, including distance-based validation methods, have been widely investigated in remote sensing, geographic information systems, and location-based services [7,8,30–35]. Despite their effectiveness, these techniques are rarely integrated with AI-assisted visual validation as an operational constraint within waste-management workflows. GreenSnap-AI bridges this gap by combining AI-based image verification, geo-fenced cleanup validation, and role-based civic monitoring within a unified software framework.

Existing smart waste-management systems typically address isolated functionalities, including image-based waste recognition, citizen complaint reporting, environmental mapping, or location-aware monitoring. However, most existing solutions do not integrate these capabilities into an enforceable cleanup-verification workflow. Table 1 compares representative waste detection and civic monitoring approaches based on visual validation, real-time operation, geospatial verification, enforcement capability, and citizen involvement. Unlike existing systems that primarily provide detection or reporting support,

**Table 1**
Comparison of existing waste reporting systems and GreenSnap-AI across vision, real-time enforcement, and citizen participation.

| Paper | Vision only / Vision+Spatial | Offline / Real-time | Geo-verification | Cleanup Enforcement | Detection / Enforcement | Citizen-in-loop | Limitations |
|---|---|---|---|---|---|---|---|
| Zhao *et al.* [36] | Vision only | Offline | No | No | Detection | No | Focuses only on image classification |
| Han *et al.* [37] | Vision only | Real-time | No | No | Detection | Partial | Detection without spatial accountability |
| Waste-YOLO [10] | Vision only | Real-time | No | No | Detection | No | No civic reporting workflow |
| Non-profit mapping [23] | Vision only | Offline | Yes | No | Detection | Yes | Mapping-oriented, lacks AI validation |
| DRSE-YOLO [20] | Vision only | Real-time | No | No | Detection | No | Focuses on detection accuracy without civic workflow |
| MRS-YOLO [21] | Vision only | Real-time | Yes | No | Detection | No | Improves detection precision but lacks location verification |
| FE-YOLO [22] | Vision only | Real-time | No | No | Detection | No | Optimized for waste detection without spatial accountability |
| **GreenSnap-AI (this work)** | Vision + Spatial | Real-time | Yes | Yes | Enforcement | Yes | Unified geo-verified AI workflow |

GreenSnap-AI combines AI-assisted image validation, geo-fenced verification, and role-based workflow management to ensure that cleanup activities are both visually confirmed and geographically validated.

By requiring both visual evidence and spatial proximity before report resolution, the platform provides a more reliable, transparent, and accountable approach to municipal waste-management operations.

## 2. Software description

GreenSnap-AI is a full-stack, open-source software platform designed to support end-to-end waste reporting, AI-assisted verification, supervisor cleanup validation, and administrative monitoring. The system was developed using a modular service-oriented architecture. The mobile application, web dashboard, backend API, database, and AI inference services operate as independently maintainable components. This design improves scalability, flexibility, and long-term maintainability while allowing AI models and frontend interfaces to evolve without major changes to the core application logic.

GreenSnap-AI is implemented as a modular software ecosystem consisting of mobile clients, a web-based administrative dashboard, backend API services, cloud-based AI inference modules, and a geospatial monitoring layer. Detailed architectural components and implementation details are described in Sections 2.1 and 2.2. To ensure accountability and prevent fraudulent or off-site waste resolution, GreenSnap-AI enforces a closed-loop decision workflow that tightly integrates visual AI validation with mandatory geospatial verification. Unlike vision-only reporting systems, both citizen-submitted reports and supervisor cleanup actions must satisfy enforced AI and location-based constraints before a report can be finalized. Fig. 1 illustrates the complete geo-verified AI decision loop implemented in the system.

Fig. 1 illustrates the geo-verified workflow implemented in GreenSnap-AI. Waste reports are validated using AI inference services, while cleanup requests undergo both visual re-validation and Haversine-based geo-fence verification. A report is marked as resolved only when the submitted cleanup evidence satisfies both visual and spatial validation requirements, ensuring accountability and reliable waste-management operations.

### 2.1. Software design

GreenSnap-AI is implemented as a modular service-oriented system following the Model–View–Controller (MVC) architectural pattern. The software architecture was designed to support scalability, maintainability, interoperability, and independent component evolution. Four primary design principles guided the implementation: (i) modularity, enabling the mobile applications, backend services, and AI inference modules to evolve independently; (ii) model-agnostic integration, allowing visual inference models to be replaced or upgraded without modifying the core business logic; (iii) configurable validation policies, supporting adjustable confidence thresholds and geo-fence constraints; and (iv) auditability, ensuring that reporting, inference, and cleanup-validation actions remain traceable. These principles support long-term extensibility and facilitate deployment across different municipal or environmental monitoring scenarios.

The selected architectural principles enable independent evolution of user interfaces, backend services, data management, and AI inference components.

Security and operational requirements are addressed using established web-development practices. Role-based access control is implemented using JWT-based authentication, while user credentials are protected through bcrypt password hashing. All communication between clients and backend services is secured through HTTPS, and Cross-Origin Resource Sharing (CORS) policies are enforced to restrict unauthorized access. The administrative dashboard uses Leaflet.js with OpenStreetMap tiles for visualization of geo-tagged reports and cleanup activities. Administrators can monitor report status, supervisor activity, worker assignments, and operational statistics through interactive dashboards and filtering tools. Overall, the software architecture emphasizes modularity, traceability, maintainability, and independent evolution of system components.

The modular architecture enables adaptation to other geo-referenced civic-monitoring applications by replacing only the task-specific visual inference component.

### 2.2. Software architecture

GreenSnap-AI follows a service-oriented MVC architecture that separates data management, business logic, user interaction, and AI inference into independent components. The Model layer uses MongoDB with Mongoose schemas for storing user, report, image, and geospatial metadata, while the Controller layer is implemented through Node.js/Express APIs responsible for authentication, workflow management, AI-service coordination, and geo-verification. The View layer consists of the React Native mobile application and React.js administrative dashboard. AI inference tasks are delegated to external services to improve scalability and maintainability.

Key backend routes include /submitReport, /submitCleanup, and /getReports. During report submission, the backend first validates the authentication token and request payload. The submitted image is then uploaded to cloud storage and forwarded to the configured AI inference service. The returned classification confidence is evaluated before storing the report. If the image is classified as waste above the configured threshold, the report is stored in MongoDB with a pending status; otherwise, the submission is rejected. During cleanup verification, the backend validates the cleanup image and computes the geographical distance between the original report coordinates and the cleanup

coordinates using the Haversine formula. If the calculated distance is within the predefined geo-fence threshold (10 m), the report status can be updated to resolved; otherwise, the cleanup request is flagged for administrative review or rejection.

AI inference is performed through external RESTful services rather than within the backend server, improving scalability, maintainability, and deployment flexibility. This modular design enables independent upgrading or replacement of inference models without modifying the core application workflow as presented in Fig. 2.

### 2.3. Software functionalities

GreenSnap-AI provides integrated modules for waste reporting, AI-assisted verification, supervisor validation, administrative monitoring, and operational analytics, supporting practical municipal workflows across mobile and web interfaces.

- Report Creation: Citizens can capture or upload waste images through the mobile application. The system automatically attaches geospatial coordinates using the device location service. Before a report is stored, the submitted image is forwarded to the configured AI inference service for validation. If the returned confidence exceeds the predefined threshold and the image is classified as waste, the report is accepted and stored in the database; otherwise, the user is prompted to resubmit a valid image. This process reduces non-waste or fraudulent submissions and improves reporting reliability.
- Supervisor Validation: Supervisors are provided with access to assigned pending reports through the mobile application. During cleanup verification, the supervisor captures a new image and current geospatial coordinates at the cleanup location. The backend validates the submitted cleanup image using the AI inference service and computes the geographical distance between the original report coordinates and the cleanup coordinates using the Haversine formula. A cleanup request is considered valid when the calculated distance falls within the configured geo-fence threshold. In the current implementation, a 10-meter threshold was used during field testing to balance location accuracy and operational accountability.
- The geo-verification module calculates the distance between the original report location and cleanup location using the Haversine formula. The backend accepts a cleanup request only when the AI confidence score satisfies the predefined threshold and the calculated distance remains within the configured 10-meter geo-fence. Otherwise, the request is rejected or forwarded for administrative review. The complete validation logic is summarized in Algorithm 1.

- Administrative Dashboard: The administrative web dashboard enables centralized monitoring and management of submitted reports, cleanup activities, users, and operational analytics. Reports are visualized on interactive Leaflet.js maps using OpenStreetMap tiles and are color-coded according to their current status (e.g., pending, in-progress, resolved, or rejected). Administrators can inspect report details, review associated images, assign supervisors, monitor cleanup progress, and analyze operational statistics through interactive charts and filtering tools. The dashboard also provides summary analytics, including report trends, user engagement activity, supervisor performance, and report-status distributions.
- User Management: The platform implements role-based access control to manage citizens, supervisors, workers, and administrative users. Administrators can create, update, assign, or remove user accounts through the dashboard interface. Access permissions are enforced according to user roles, ensuring that only authorized users can access specific functionalities such as cleanup verification, report approval, administrative analytics, or report submission interfaces.

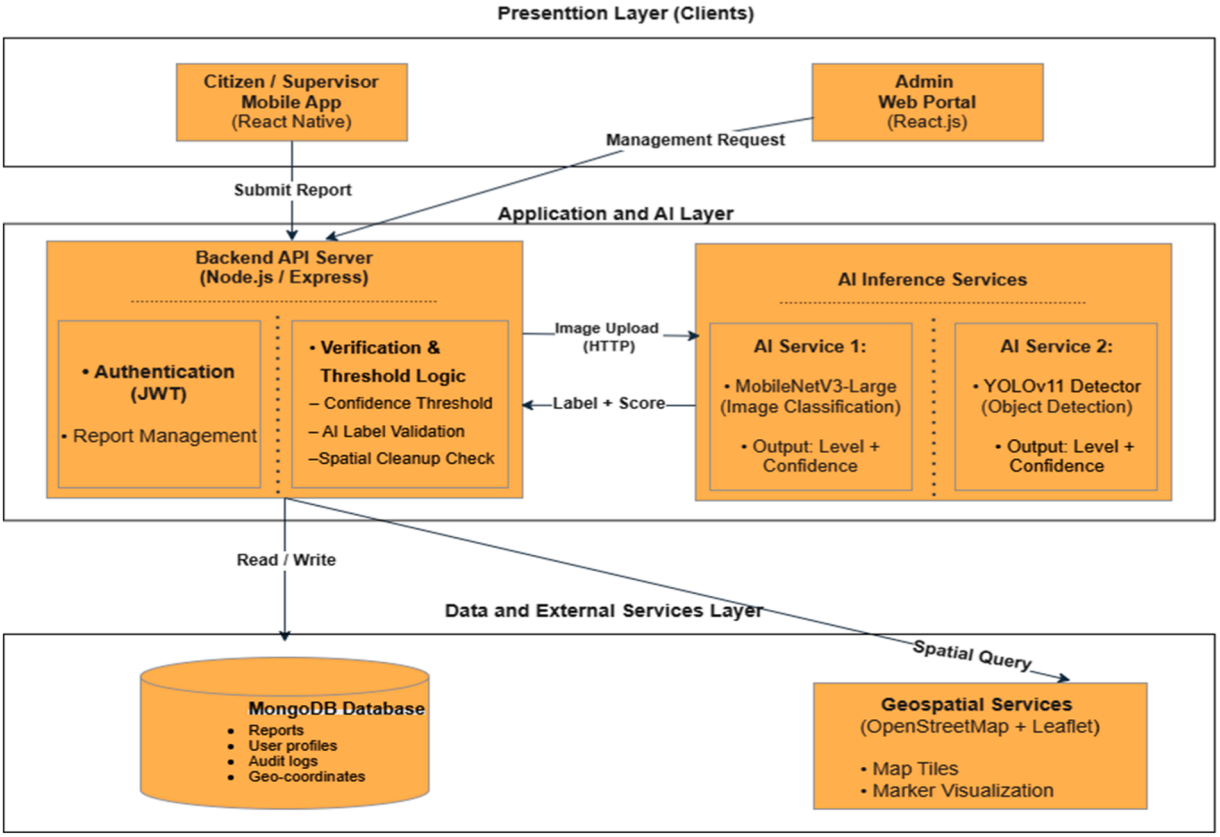



**Fig. 2.** Overall software architecture of GreenSnap-AI illustrating the interaction among React Native mobile clients, React.js administrative dashboard, Node.js/Express backend services, MongoDB database, AI inference services, cloud storage, and geospatial visualization modules.

**Algorithm 1**

Input:
Original report coordinates, cleanup coordinates, AI confidence score
1. Validate cleanup image using AI inference service.
2. Calculate geographical distance between report and cleanup locations.
3. If AI confidence satisfies threshold and distance $\leq$ 10 m:
Accept cleanup request.
Else:
Reject or flag request for review.
Output:
Cleanup validation decision.

- AI Model Monitoring: The platform records inference outputs, classification confidence scores, and validation outcomes for submitted reports and cleanup requests. These logs allow administrators to monitor model behavior, evaluate operational performance, and support future model retraining or threshold adjustment. The modular inference architecture also enables independent replacement or upgrading of AI models without requiring changes to the remaining software infrastructure.

The integration of mobile imaging, AI-assisted validation, geospatial verification, and administrative monitoring establishes a complete operational workflow for municipal waste reporting and cleanup supervision. The software interfaces were designed to minimize user complexity while supporting accountability and operational transparency. In addition to waste-management applications, the modular architecture can be adapted to other geo-referenced civic monitoring scenarios such as infrastructure damage reporting, illegal dumping detection, environmental hazard monitoring, and smart-city maintenance operations.

### 2.4. Evaluation, results and discussion

#### 2.4.1. Dataset and training

A custom binary waste-image dataset containing 17,556 labeled images was used to evaluate the visual validation component of GreenSnap-AI. The dataset was constructed by combining publicly available waste image samples from the LitterR dataset with additional real-world waste photographs collected to represent practical deployment conditions. The collected images covered diverse environmental scenarios, including variations in illumination, background complexity, and object appearance. Each image was annotated into one of two operational classes: waste and non-waste. The final held-out test subset contained 1755 images, including 885 waste samples and 870 non-waste samples, providing a balanced evaluation setting for binary waste verification. The class labels were manually reviewed to ensure consistency before model training and testing. The annotation process was designed according to the application objective of identifying valid waste-reporting events. The task focused on binary waste verification rather than fine-grained waste-category recognition. Although the dataset includes variations in illumination, background complexity, and object appearance, it represents a binary verification scenario designed for determining whether a submitted image contains actionable waste evidence. Therefore, the reported performance should be interpreted within this operational context rather than as a generalized multi-category waste recognition capability.

Images were manually reviewed to verify label consistency and remove unsuitable samples before model training. The final dataset was divided into training, validation, and held-out test subsets using the prepared labels for model development and unbiased evaluation.

For YOLOv11 training, the model was configured as a waste-presence verification detector, where the objective was to identify whether actionable waste evidence existed within a submitted image. The model output was used as a confidence-based validation signal rather than as a fine-grained waste classification mechanism. This formulation aligns with the operational objective of determining whether a submitted report contains valid waste content rather than performing fine-grained object counting or segmentation. No external benchmark dataset was used during evaluation; therefore, all reported performance metrics correspond to the held-out test subset derived from the collected dataset. The dataset was divided into training, validation, and held-out test subsets for evaluation. The binary classification task included waste and non-waste categories, and the test subset was reserved exclusively for final performance assessment. Data augmentation, including random horizontal flipping and controlled zoom operations, was applied only during training to improve generalization and reduce overfitting risk. Future work will investigate cross-dataset validation and broader environmental conditions.*Inference Integration and Classification Result:*

After training, both visual inference models were deployed as independent RESTful AI services to support scalable integration with the GreenSnap-AI backend infrastructure. The MobileNetV3-Large model was hosted through Hugging Face Spaces using a Gradio-based interface, while YOLOv11 inference was deployed through the Ultralytics inference service. During report submission, the backend forwards uploaded images to the configured inference endpoint and receives the predicted label and confidence score in real time. This decoupled deployment strategy allows AI inference to operate independently from the core backend services, improving scalability, maintainability, and flexibility for future model replacement or upgrades.

The MobileNetV3-Large model was trained using a transfer-learning strategy based on ImageNet pre-trained weights. Instead of training the network from scratch, the pre-trained backbone was utilized to leverage previously learned low-level visual representations, reducing computational cost and improving convergence efficiency for the binary waste-classification task. The feature-extraction layers of the backbone network were initially frozen, while a custom classification head was appended consisting of a GlobalAveragePooling2D layer followed by a fully connected dense layer with sigmoid activation for binary prediction. Dataset loading, preprocessing, and augmentation were implemented using TensorFlow Keras data generators. On-the-fly augmentation techniques including random flipping and controlled zoom operations were applied during training to improve generalization and reduce overfitting.

The training curves shown in Figs. 3 and 4 demonstrate stable convergence behavior for both models throughout the training process. Validation accuracy increased consistently alongside training accuracy, while training and validation losses progressively decreased, indicating effective learning with limited evidence of overfitting. MobileNetV3-Large achieved strong discrimination capability, while YOLOv11 demonstrated superior convergence characteristics and classification stability on the held-out test subset. Receiver Operating Characteristic (ROC) analysis further confirmed the effectiveness of both models, with YOLOv11 demonstrating superior discrimination capability across the evaluated test samples.(see Fig. 5)

Model performance was evaluated using the held-out test subset. YOLOv11 achieved exceptionally strong waste-presence verification performance, correctly identifying 879 waste samples and 867 non-waste samples, with only 3 false positives and 6 false negatives. These results correspond to an overall accuracy of approximately 99.5%, demonstrating highly reliable waste-presence verification capability

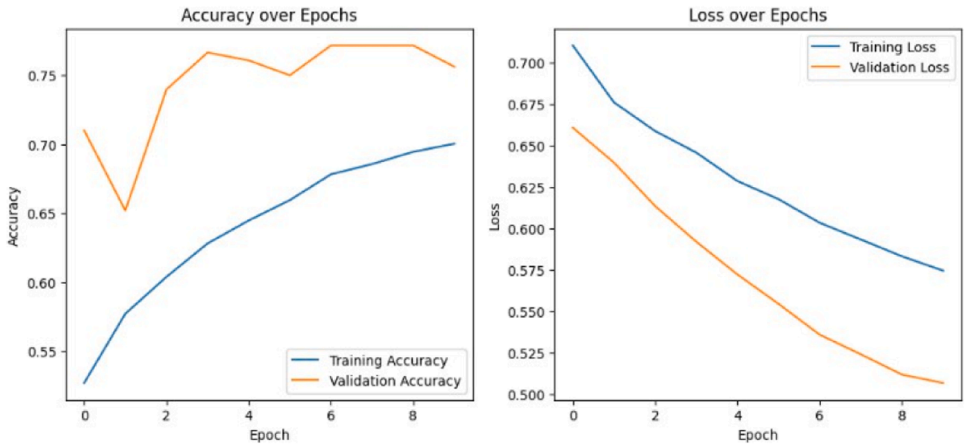



**Fig. 3.** Training and validation performance of the MobileNetV3-Large model showing accuracy improvement and loss reduction across training epochs for binary waste classification.

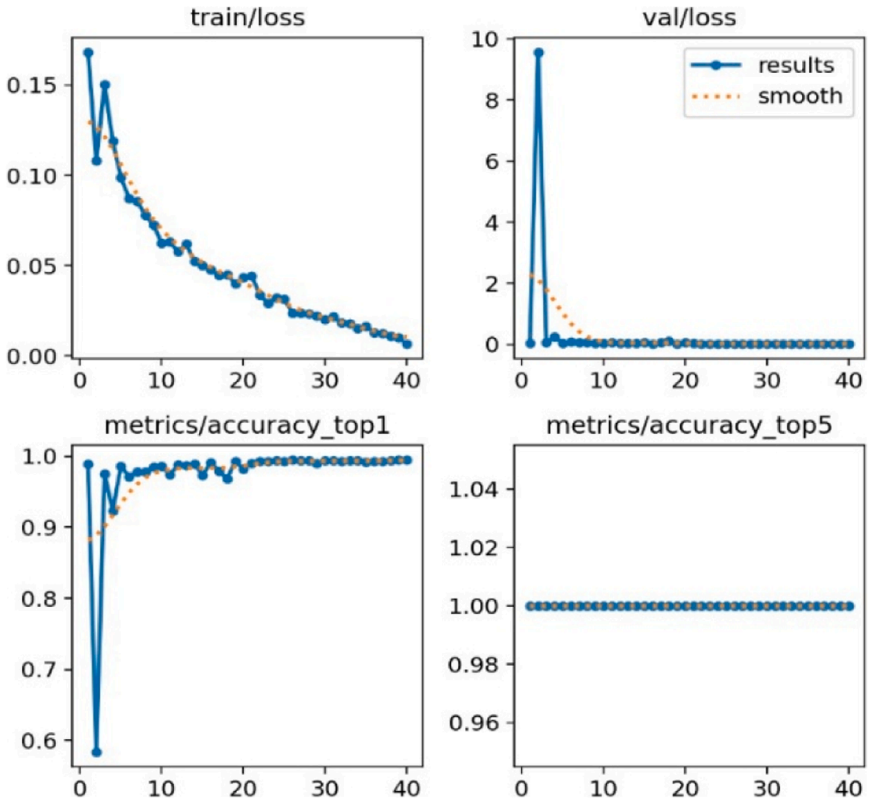



**Fig. 4.** Training convergence behavior of the YOLOv11 model showing accuracy progression and loss variation across epochs during binary waste-image classification.

within the evaluated dataset as presented in Table 3. In comparison, the MobileNetV3-Large classifier achieved 817 true positives and 807 true negatives, with 64 false positives and 69 false negatives, resulting in an overall accuracy of approximately 92.4%. The confusion matrices presented in Fig. 6 further illustrate the superior classification consistency achieved by the YOLOv11-based inference service.

Beyond classification accuracy, the software evaluation considered operational performance of the integrated inference workflow. During deployment testing, image submission, backend processing, AI inference communication, and result storage were evaluated as an end-to-end pipeline. The cloud-based inference services introduced an average response latency of approximately 1–2 s per report submission, depending on network conditions and external service response time. The modular deployment architecture enabled independent execution of AI inference services without affecting the responsiveness of the core backend application. Table 2 presents the comparison of the prior work.

From an operational perspective, the integrated mobile application and backend infrastructure demonstrated stable real-time performance during field testing. AI inference requests introduced an average latency of approximately 1 - 2 s per report submission, depending on network conditions and cloud-service response times. The React.js administrative dashboard rendered geospatial maps, report information, and analytical summaries efficiently within standard desktop browser environments. Informal testing with local volunteers indicated that the reporting workflow was responsive and easy to use under typical operating conditions.

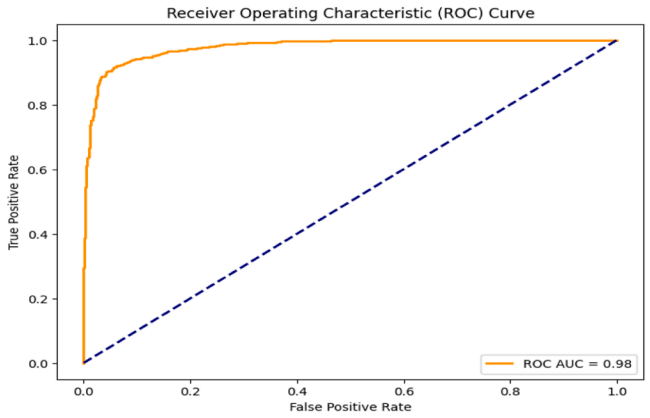



**Fig. 5.** Receiver Operating Characteristic (ROC) curve of the MobileNetV3-Large model demonstrating classification discrimination capability on the held-out test subset.

### 2.4.2. Statistical calculation

Standard binary-classification metrics were computed to evaluate model performance, including accuracy, precision, recall, true positive rate (TPR), true negative rate (TNR), and Receiver Operating Characteristic Area Under Curve (ROC-AUC). Accuracy was computed as the ratio of correctly classified samples to the total number of evaluated samples. Precision and recall were calculated to assess prediction reliability and detection sensitivity, respectively. For YOLOv11, the model achieved a true positive rate of approximately 99% and a true negative rate approaching 100%, indicating strong discrimination capability for both waste and non-waste categories. Confidence intervals were estimated using standard binomial approximations, with observed variations remaining within ±0.5% due to the relatively large evaluation subset. Since the geospatial verification stage relied on deterministic Haversine-distance calculations with predefined thresholds, no stochastic or probabilistic modeling was required for spatial validation.

### 2.4.3. Field testing and discussion

GreenSnap-AI was deployed experimentally in Sagri, Rawalpindi, to assess practical feasibility and operational performance. Municipal personnel used the platform for waste reporting, cleanup validation, and administrative monitoring. The deployed system demonstrated responsive interaction between mobile clients, backend services, AI inference modules, and the administrative dashboard. AI inference requests required approximately 1–2 s per report submission depending on network conditions and cloud-service response time. The service-oriented architecture allowed AI models to operate independently from the backend workflow, supporting flexible deployment and future scalability. A comprehensive stress test under large-scale concurrent municipal usage was beyond the scope of this initial deployment and remains a direction for future evaluation.

### 2.4.4. Limitations

Several practical limitations were identified during experimental deployment. Real-world waste images captured under poor lighting conditions, motion blur, partial occlusion, or environmental clutter may reduce classification reliability compared with curated evaluation datasets. In addition, GPS uncertainty caused by device accuracy, environmental conditions, and signal availability may affect geo-fence validation outcomes. The current 10-meter threshold was selected as an initial operational parameter for field testing; however, adaptive thresholds based on positioning uncertainty and deployment-specific calibration may further improve robustness in large-scale deployments. The current implementation focuses on binary waste-presence verification and does not distinguish between specific waste categories such as plastic, organic, or recyclable materials. Consequently, the reported accuracy should be interpreted within this limited operational scope and not as comprehensive waste recognition performance. Future work will investigate larger and more diverse datasets

**Table 2**
Comparative waste-presence verification performance of the proposed GreenSnap-AI models.

| Proposed models | Accuracy |
|---|---|
| YOLOv11 waste-presence verification | 99.5% accuracy. |
| MobileNetV3-Large | 92.43% accuracy. |

**Table 3**
Comparison of GreenSnap-AI performance with representative waste-recognition approaches reported in the literature.

| Prior Work | Accuracy |
|---|---|
| Zhao *et al.* (MobileNetV3-Large, (multi-class) | 81% accuracy. |
| Han *et al.* (YOLOv3- MobileNetV3_Lite, 4 classes): | 94.6% accuracy. |

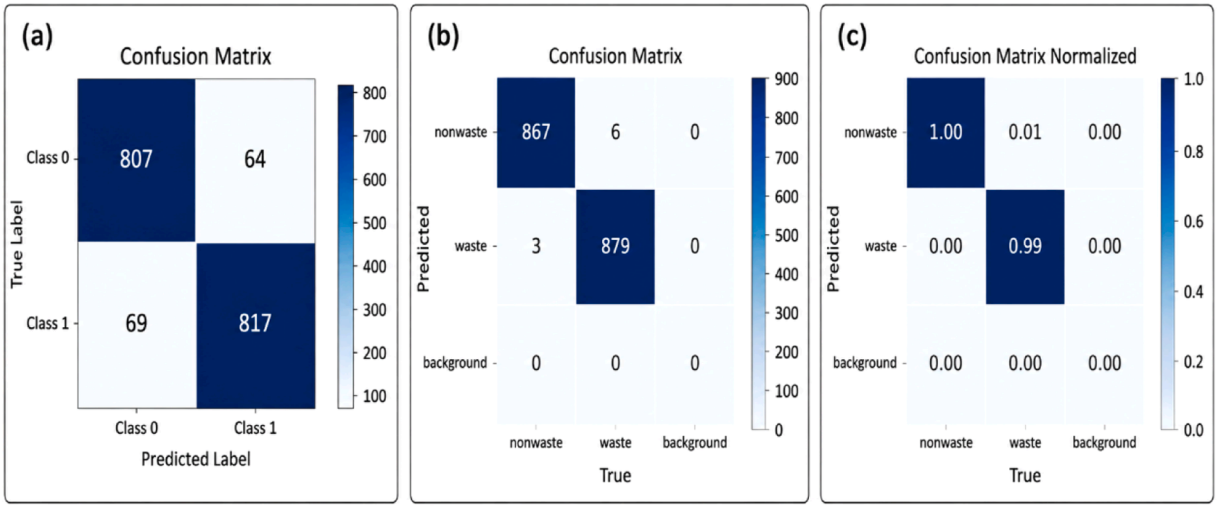



**Fig. 6.** Confusion matrix-based evaluation of GreenSnap-AI visual validation models on the held-out test subset: (a) MobileNetV3-Large classification results; (b) Raw YOLOv11 prediction counts for waste and non-waste classes; (c) Normalized YOLOv11 confusion matrix showing class-wise prediction performance and misclassification distribution.

covering multiple waste categories, geographical regions, and environmental conditions. Furthermore, the present system depends on internet connectivity for cloud-based AI inference services, which may affect usability in low-connectivity environments. Although the modular architecture supports scalability, large-scale stress testing and comprehensive security evaluation were beyond the scope of the current deployment and remain important directions for future investigation.

The current YOLOv11 implementation is designed for binary waste-presence verification within the reporting workflow rather than detailed object localization or multi-class waste categorization. Future extensions may incorporate instance-level detection and segmentation approaches for identifying specific waste types and quantities.

#### 2.4.5. Discussion

The geo-fence parameter is configurable and can be adjusted according to deployment conditions, GPS accuracy, and municipal operational requirements. The YOLOv11-based inference service achieved substantially stronger waste-presence verification performance than the MobileNetV3-Large baseline within the evaluated dataset. Beyond classification accuracy, the primary contribution of GreenSnap-AI is its end-to-end software workflow. The platform combines mobile reporting, AI inference, geospatial validation, administrative monitoring, and role-based operational management within a unified system. The modular microservice architecture further improves extensibility by enabling independent replacement or upgrading of inference models without affecting the remaining system components. Broader deployment will require evaluation under more diverse environmental conditions and larger operational scales.

#### 2.4.6. Participants details

Field testing involved municipal personnel and volunteer citizens who interacted with the platform for waste reporting, cleanup verification, and administrative monitoring activities.

#### 2.4.7. Procedure

Participants used the mobile and web applications during routine waste-reporting and cleanup-verification activities. Report submissions, classification outcomes, and geospatial validation results were automatically recorded by the backend infrastructure. Informal feedback indicated positive usability and operational responsiveness during field deployment.

## 3. Impact

GreenSnap-AI provides a reusable software framework for AI-assisted civic reporting and geo-verified task validation. Beyond waste management, the modular architecture can be adapted to other geo-referenced monitoring applications such as infrastructure reporting, environmental monitoring, and smart-city services. The platform supports transparency, accountability, and extensibility through its service-oriented design.

## 4. Conclusion

This paper presented GreenSnap-AI, a full-stack software platform for AI-assisted and geo-verified municipal waste reporting and cleanup validation. The proposed system integrates AI-assisted waste verification with geo-enforced cleanup validation within a unified municipal reporting workflow. Experimental evaluation demonstrated strong classification performance for waste verification, while field testing confirmed the practical feasibility of integrating AI-assisted validation with Haversine-based geospatial enforcement in real-world municipal reporting scenarios.

The primary contribution of GreenSnap-AI is the development of a reusable software workflow rather than a new AI algorithm. The platform integrates existing AI inference capabilities, geospatial verification, citizen reporting, and administrative monitoring into a scalable architecture that enables transparent and accountable waste-management operations.

Future work will focus on extending the platform with multi-class waste categorization, predictive analytics for waste-accumulation forecasting, offline on-device inference using lightweight edge models, and broader deployment across larger municipal environments. Additional scalability testing, security evaluation, and integration with smart-city GIS infrastructures will also be investigated to support wider operational adoption.

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## CRediT authorship contribution statement

**Mohammed Ali:** Software, Methodology, Formal analysis, Data curation, Conceptualization. **Hela Elmannai:** Supervision, Project administration, Investigation, Funding acquisition, Data curation. **Abid Jameel:** Writing – review & editing, Writing – original draft, Validation, Supervision, Formal analysis, Conceptualization. **Abeer D. Algarni:** Visualization, Resources, Methodology, Funding acquisition. **Dina Abed Alqahtani:** Methodology, Investigation, Formal analysis, Conceptualization. **Iftikhar Ali Khan:** Resources, Project administration, Investigation, Formal analysis, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests, personal relationships, institutional affiliations, or commercial associations that could have appeared to influence the work reported in this paper.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.softx.2026.102964.

## Data availability

The waste image dataset and source code are publicly available in the linked repository (see Metadata). All user-related data collected in field tests were anonymized. Additional logs and analytics summaries are available upon request to the corresponding author.

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